Introduction

- Objective: Detect moss areas on wall surfaces.
- Applications: Maintenance, restoration, and ecological monitoring.
- Challenges:
- - High visual similarity with background
- - Lighting variations and color inconsistency
- - Complex textures

The goal of this project is to detect and segment **moss regions on wall surfaces**, mainly focusing on their greenish color and texture patterns. The main objective is to **maximize the F-measure** value. This presentation explores two distinct methods for vegetation detection: a direct approach and a modern deep learning technique as indirect approach. We will look into their methodologies, evaluate their performance, and compare their strengths and limitations.

Method 1: Hybrid (Edge Detection + Active Contour)

- This non-learning approach combines image processing and contour refinement. It involves preprocessing, edge detection, region initialization, and active contour refinement.

Preprocessing
- Green channel extraction
- Contrast enhancement (CLAHE)
- Noise reduction (Median filtering)

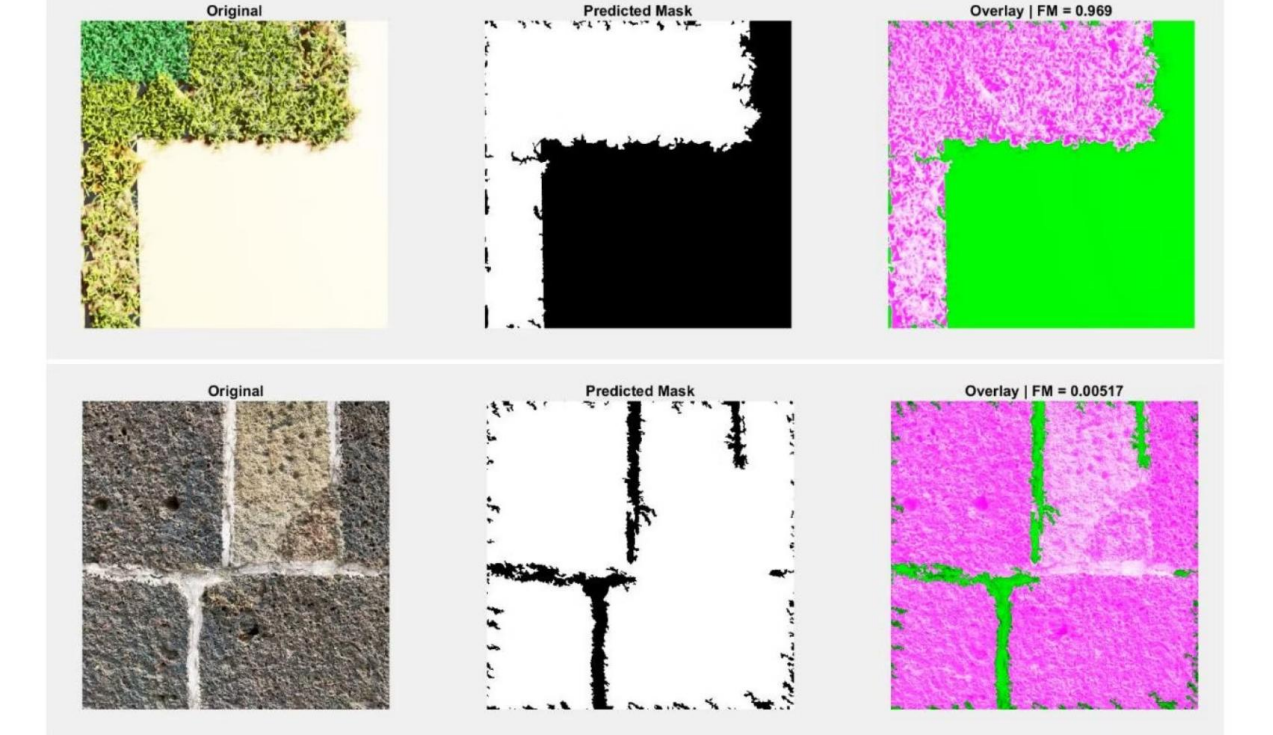
Edge Detection & Initialization
- Canny edge detection
- Morphological closing
- Hole filling & small region removal

Region Refinement
- Chan-Vese active contour algorithm (up to 100 iterations)
- Final hole filling & small-region filtering

This method excels with high contrast and well-separated vegetation but struggles with cluttered textures or noisy patterns. Evaluation uses FM, CR, and CM metrics.

Approach 1 Procedure

Best case and Worst case for direct approach

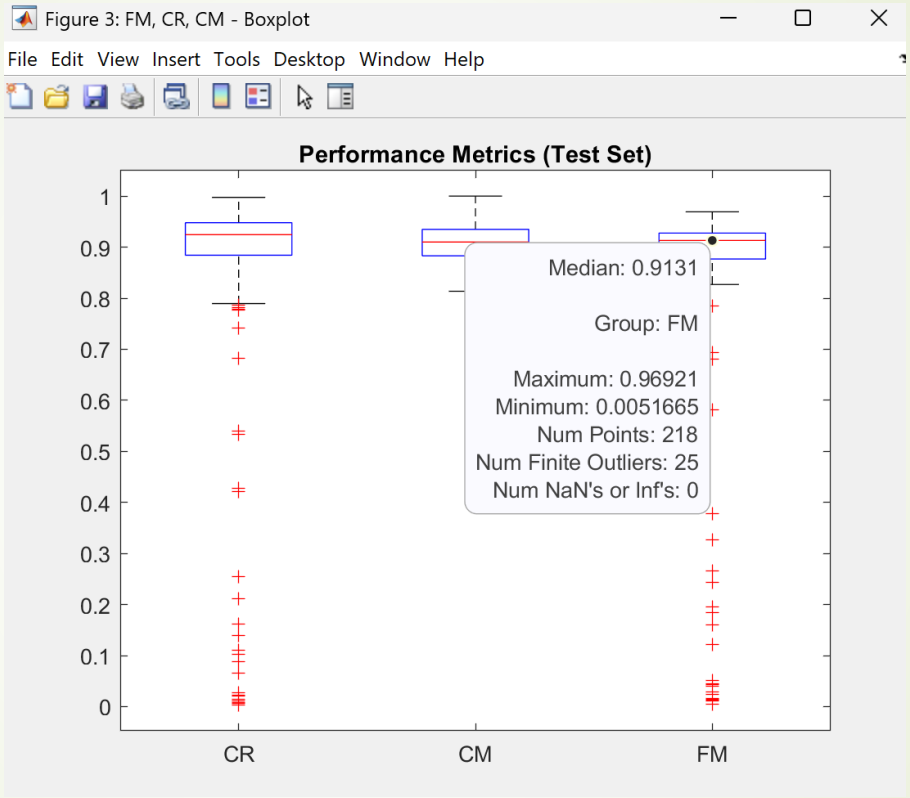


- Clear vegetation patch with high contrast.
- Accurate outline, minimal noise.
- Strong contrast, uniform texture, low noise.

- Wall with stone/brick texture, irregular mortar lines.
- Mistakenly detects texture as vegetation.
- Wall texture resembles edge patterns, noisy background.

Approach 1 Procedure

Best case and Worst case for direct approach



Final average: CR: 0.833 , CM: 0.902 , FM: 0.831

High CM (0.902) means most of the actual vegetation is correctly captured ⇒ low under-segmentation.
Lower CR (0.833) indicates that some predicted regions are incorrect ⇒ presence of false positives.
The model tends slightly toward **over-segmentation** — it detects most of the vegetation but also includes some extra, incorrect regions.

Direct Method: Pros & Cons

Advantages:

- No training required – Works out of the box without labeled data.
- Fast and lightweight – Suitable for systems without GPU or limited computing power.
- Interpretable pipeline – Each step is transparent and manually tunable.
- Performs well on high-contrast, clean images – Especially when vegetation stands out clearly.

Limitations:

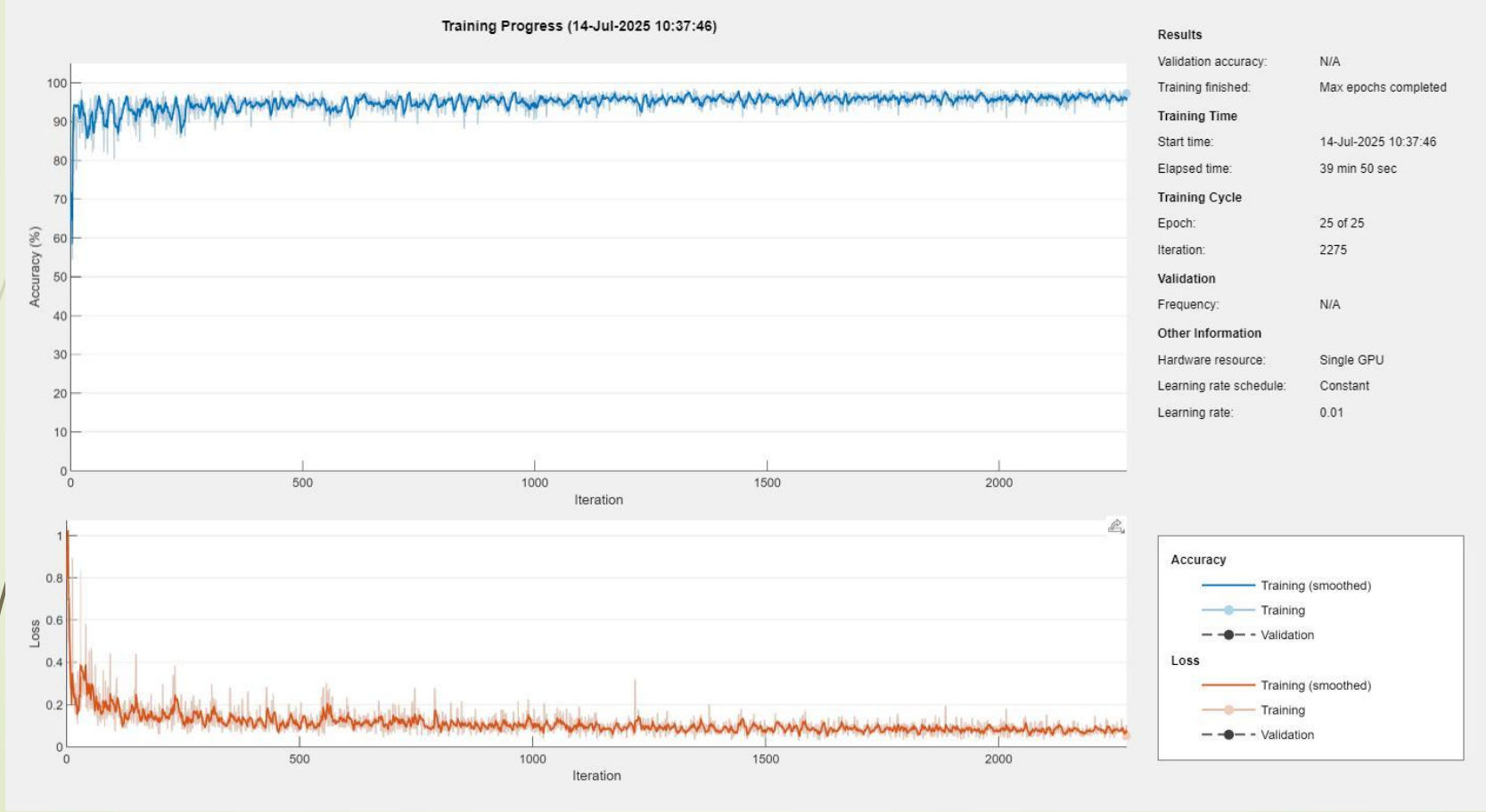
- Not adaptive – Fixed rules make it sensitive to texture or lighting variations.
- Poor generalization – Struggles with complex or noisy backgrounds (e.g. bricks, shadows).
- High false positives in textured areas – Edge detector is easily misled by patterns.
- Fails on cluttered scenes – Accuracy drops sharply on low-contrast or occluded vegetation.

Approach 2: Deep Learning

- The deep learning approach in our project utilizes **DeepLabv3+ with a ResNet-50 backbone** for semantic segmentation.
  The model is trained on RGB wall images and their corresponding moss masks.
  We apply **data augmentation techniques** such as rotation and flipping to improve generalization and balance the class distribution using **weighted loss functions**.
- After training for 25 epochs, the model produces **pixel-level segmentation masks** for moss areas.
  Post-processing is applied to refine the predictions, including **hole filling**, **morphological filtering**, and **largest connected component selection**.
- Evaluation is done on both training and test sets using FM, CR, and CM metrics.

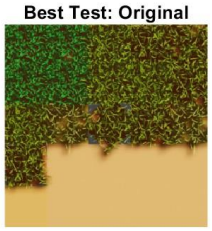
Results

Deep Learning Training



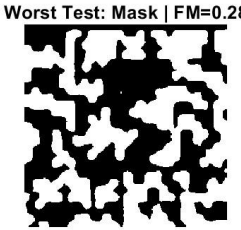
Results

Deep Learning
Best and Worst cases





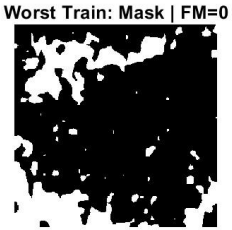






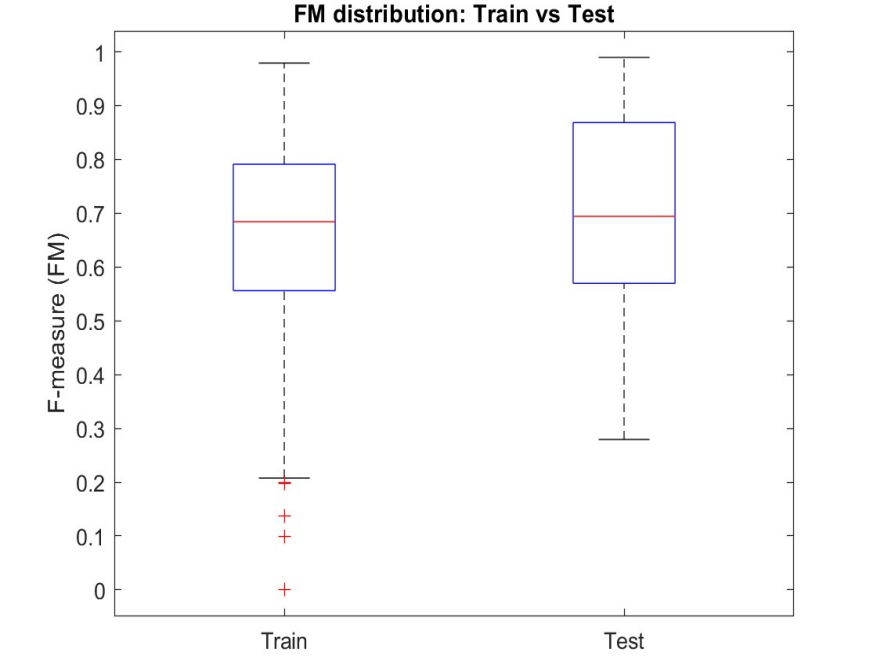






Results

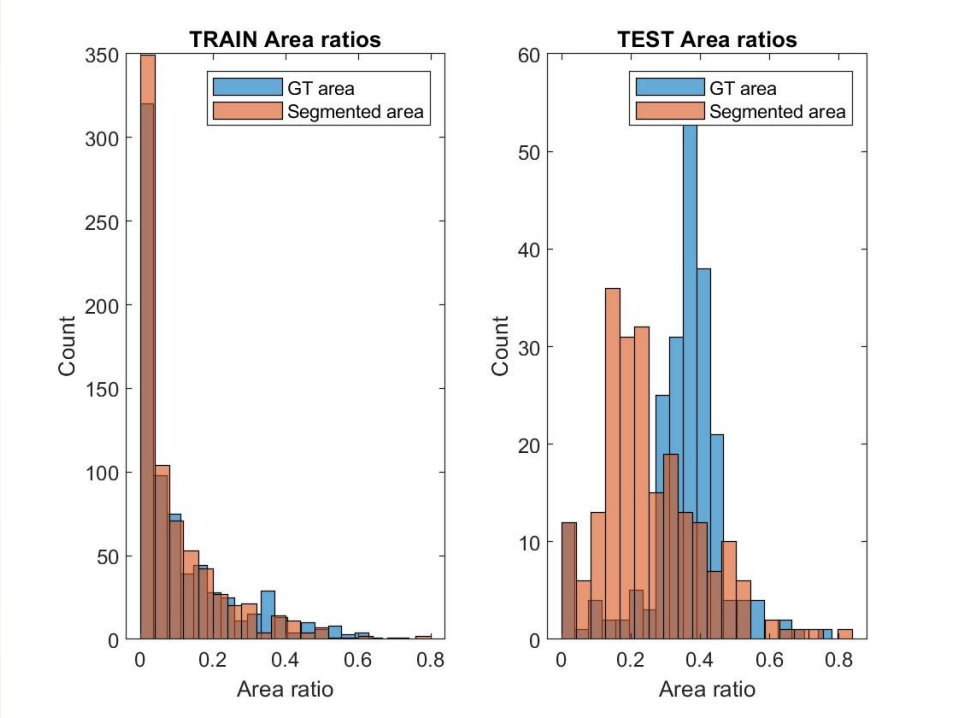
Deep Learning
Boxplot and Histogram



- Median FM is slightly higher on the training set, but the difference is small → good generalization.
- Outliers in both sets show that the model struggles on some difficult images.
- No clear signs of overfitting.

Results

Deep Learning
Boxplot and Histogram



- In both Train and Test sets, predicted segmented areas are often smaller than ground truth.
- This indicates a tendency toward under-segmentation, especially on test data.

DeepLabv3+: Pros & Cons

- Advantages:
- - High accuracy and robustness
- - Learns from data directly (features, textures)

- Limitations:
- - Requires large labeled dataset
- - Computationally intensive

Conclusion

Comparison Summary: Hybrid vs DeepLabv3+

- The **Hybrid method** is fast, simple, and works well on **high-contrast images**, but it's **not adaptable**. It often **over-segments** on textured backgrounds like bricks or noisy walls.
- The **DeepLabv3+ method** is more powerful and **learns complex patterns**. It performs better on **diverse and real-world images**, but needs **labeled data** and may **under-segment** in some cases, especially when the training data is limited.

Thank you for your attention!